

(72 points possible)

✓ Problem 1: Getting started (12 points [2,2,4,4])

a, 2 pts) Write an expression that evaluates to the ninth power of 8.

```
# TODO
eightPowNine=8**9
```

b, 2 points) Write an expression that evaluates to a string that says "the tenth power of 8:" followed by the tenth power of 8 (cast to a string and use concatenation).

```
# TODO
eightPowTenStr= "the tenth power of 8:" , 8**9
```

c, 4 points) Write a short program that asks the user for a number, then prints that power of 8 (any other text in the output is optional).

```
# TODO
print(8**int(input("Enter a number.")))
```

d, 4 points) Write a program which prompts the user for a string, like, "Enter a string:" On getting this string, the program should determine whether your string was strictly larger than 10 characters. Depending on this condition, your program should print either "[X] was my string and is longer than 10 characters" or "[X] was my string and is less than or equal to 10 characters" (where [X] is replaced by the input string).

```
#TODO
word = input("Enter a string:")
if (len(word)>10):
    res = "longer than"
else:
    res = "less than or equal to"
print(f"{word} was my string and is {res} 10 characters")
```

✓ Problem 2: Types (8 points [1x8])

Each code box below contains a literal value (i.e., not a variable). In the text box below the code box, give its type. Your options are *integer*, *boolean*, *float*, or *string*.

Fun fact: "foo" is a commonly used nonsense word in programming when something just needs an arbitrary name; other such words are "bar" and "baz".

```
'foo'
#string
```

a)

```
143
#integer
```

b)

```
False
#boolean
```

c)

```
19.3
#float
```

d)

Next, try to guess what type (from the four we already talked about) would be most likely to be contained in each of the following variables, given their names. Each type is used once.

is_blue

#boolean

e)

stock_price

#float

f)

student_name

#string

g)

class_count

#integer

h)

#there is no question????

✓

Problem 3: Comparisons and conditions (16 points
[3,7,6])

Write a brief program for each prompt that makes use of the provided variable `current_hour` to determine what hour of the day it is. `current_hour` can be set to any value between 0 (midnight) and 23 (11pm).

```
current_hour = 10 # change to get different outputs below
```

a, 3 pts) If the current hour is between 5 and 9 inclusive, print "Breakfast!". If the current hour is between 10 and 13 inclusive, print "Brunch!". Otherwise, print "Not breakfast :(".

```
#TODO
if(current_hour>=5 and current_hour<=9):
    print("Breakfast!")
elif(current_hour>=10 and current_hour<=13):
    print("Brunch!")
else:
    print("Not Breakfast :(")
```

b, 6 points) If the time is equal to or after 9:00pm (21), print "In three hours, it will be tomorrow." Otherwise, print "In three hours, it will be " followed by the hour three hours later using non-military time (so between 1 and 12). (For example, if `current_hour` were 16, write "In three hours, it will be 7.")

```
#TODO
if(current_hour>=21):
    print("In three hours, it will be tomorrow")
else:
    print(f"In three hours, it will be {current_hour+3-12 if current_hou
#Add the comment below before the "." in the formatted string if you
#{ "pm" if current_hour+3>12 else "am" }
```

c, 6 points) Print "X hours until midnight" if the current hour is after noon, where X is the correct number of hours, and "X hours until noon" if it's before noon. If it's noon or midnight, print "Noon" or "Midnight", respectively. (You don't need to change "hours" to "hour" if there's just one.)

```

#TODO

#Why have this:

if(current_hour==0):
    response = "Midnight"
elif(current_hour==12):
    response="Noon"
else:
    hoursUntil = 12-current_hour
    if(current_hour<12):
        response = " hours until noon"
    else:
        response = " hours until midnight"
    response = str(hoursUntil) + response
print(response,"\n")

#When you can have this:

print(f"{{( "Midnight"if current_hour== 0 else ("Noon" if current_hour ==

#IT WORKS
#MY FRANKESTIEN IS ALIVE!!!!

```

2 hours until noon

2 hours until noon

Problem 4: While: Countdown to disaster (8 points)

a, 7 points) Write a program where the user is prompted with "Enter the code: ". You can choose the code, which should be clearly assigned to a variable named *code*. If the user's input matches the code, print "You saved the world from disaster!" and end the program. Otherwise, print "Wrong code, you have N guesses left before disaster." where N is an integer variable that you set and prompt again; keep prompting until the correct code is entered or N is zero. If N reaches zero, print "Disaster".

```
#TODO
N=4
while(N>0):
    if(input("Enter the code:")=="1987"):
        print("You saved the world from disaster!")
        exit
    else:
        N-=1
        print(f"Wrong code, you have {N} guesses left before disaster.")
if(N>0):
    print("Disaster.")
```

✓ Problem 5: Find the min, max, and median (14 points)

This next program should prompt the user with "Enter a positive integer (-1 to quit)", getting a new number every time until the user enters -1. You can assume the input is an integer, but if the input is 0 or less, ignore it. After the program quits, print the min, max, and median values of the numbers entered on a single line. For example, if the input were 5, then 3, then 1, then 7, the output would be "min: 1, max: 7, median: 4". You may use the Python min and max functions. You must compute the median without the help of an outside module.

```

#TODO
inp=int(input("Enter a positive integer (-1 to quit)"))
nums = []
while(inp!=-1):
    if(inp>0):
        nums.append(int(inp))
    inp=int(input("Enter a positive integer (-1 to quit)"))
nums.sort()

minimum = int(min(nums))
maximum = max(nums)

if(len(nums)%2==0):
    median = (nums[len(nums)//2]+nums[1+(len(nums)//2)])/2
else:
    median = nums[(len(nums)//2)]
print(f"min: {minimum}, max: {maximum}, median: {median}")

[1, 2, 3, 4, 5]
min: 1, max: 5, median: 3

```

✓ Problem 6: Filling in a Program Design (12 points)

Your hard drive died, and a helpful technical professional restored what data she could. Unfortunately, a key piece of code in your data analysis pipeline was corrupted - you only have your original notes to go on, as represented by the comments below. Write the missing lines of code.

```
# Just some placeholder data – the real data was corrupted and we're get
# but this will let us run the code without errors.  Change the values a
temperatures = [10.0, 17.0, 36.0, 25.0, 30.0]
if temperatures !=[]:
    temperatures.sort()
    print(temperatures[-1],temperatures[-2],temperatures[-3])
    if(temperatures[-1]>35.0 or (temperatures[-1]>30.0 and temperatures[
        print("WARNING: DANGEROUS HEATS REACHED")
    else:
        print("Safe temperature")
else:
    print("ERROR 404: DATA NOT FOUND")

## MISSING:  if temperatures is an empty list, print an error message (a
## MISSING:  print the three largest measurements in descending orde
## MISSING:  if the two largest temperatures are greater than 30.0,
#            print a warning that there is dangerous heat.  Else prin
```

```
36.0 30.0 25.0
WARNING: DANGEROUS HEATS REACHED
```

✓ AI Statement (2 points)

Please put an X in the box that best matches your AI use for this homework ([X] in the markup will become a checked box). There is no penalty for using AI, but we want to understand how you're using it.

- ☒ I did not use AI to complete this assignment.
- ☐ I didn't use AI to debug or generate code, but I did use it as a Python reference.
- ☐ I didn't use AI to generate code, but I did use it to debug some code.
- ☐ I did use AI to generate some code, but not all the code for any problem.
- ☐ I used AI to solve at least one problem completely.

When you are done, submit both your .ipynb (File->Download->Download .ipynb in Google Colab) and a PDF (File->Print->Save to PDF) to Blackboard where you found this assignment. (The backup PDF helps us give you points if there's a problem with your .ipynb.)

Start coding or [generate](#) with AI.

